Machine Learning Techniques

Unit 1

Ankita Singh, Dept. of IT, AKGEC

Machine Learning Techniques (BCS055)

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able:		
CO 1	To understand the need for machine learning for various problem solving	K_1 , K_2
CO 2	To understand a wide variety of learning algorithms and how to evaluate models generated from data	K_1 , K_3
CO 3	To understand the latest trends in machine learning	K_2 , K_3
CO 4	To design appropriate machine learning algorithms and apply the algorithms to a real-world problems	K_4 , K_6
CO 5	To optimize the models learned and report on the expected accuracy that can be achieved by applying the models	K_4, K_5

DETAILED SYLLABUS		3-0-0
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION** – Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning and Data Science Vs Machine Learning;	08
II	**REGRESSION:** Linear Regression and Logistic Regression **BAYESIAN LEARNING** - Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm. **SUPPORT VECTOR MACHINE:** Introduction, Types of support vector kernel – (Linear kernel, polynomial kernel,and Gaussiankernel), Hyperplane – (Decision surface), Properties of SVM, and Issues in SVM.	08
III	**DECISION TREE LEARNING** - Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, ID-3 Algorithm, Issues in Decision tree learning. **INSTANCE-BASED LEARNING** – k-Nearest Neighbour Learning, Locally Weighted Regression, Radial basis function networks, Case-based learning.	08
IV	**ARTIFICIAL NEURAL NETWORKS** – Perceptron's, Multilayer perceptron, Gradient descent and the Delta rule, Multilayer networks, Derivation of Backpropagation Algorithm, Generalization, Unsupervised Learning – SOM Algorithm and its variant; **DEEP LEARNING** - Introduction,concept of convolutional neural network , Types of layers – (Convolutional Layers , Activation function , pooling , fully connected) , Concept of Convolution (1D and 2D) layers, Training of network, Case study of CNN for eg on Diabetic Retinopathy, Building a smart speaker, Self-deriving car etc.	08
V	**REINFORCEMENT LEARNING**–Introduction to Reinforcement Learning , Learning Task,Example of Reinforcement Learning in Practice, Learning Models for Reinforcement – (Markov Decision process , Q Learning - Q Learning function, Q Learning Algorithm), Application of Reinforcement Learning,Introduction to Deep Q Learning. **GENETIC ALGORITHMS:** Introduction, Components, GA cycle of reproduction, Crossover, Mutation, Genetic Programming, Models of Evolution and Learning, Applications.	08

Text books:

1. Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
3. Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.

Content

- Learning
- Types of Learning
- Well defined learning problems
- Designing a Learning System
- History of ML
- Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm)
- Issues in Machine Learning
- Data Science Vs Machine Learning

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How we Learn??

We humans learn by observing **patterns**, remembering **experiences**, and using **feedback** to improve.

The more we practise, the better we get — whether it’s solving math problems, recognising a friend’s voice, or cooking a favourite dish.

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Can Computer Learn??

Ever since computers were invented, we have wondered whether they might be made to learn.

If we could understand how to program them to learn-to improve automatically with experience-the impact would be dramatic.

Learning

A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P**, if its performance at tasks T, as measured by P, improves with experience E.

For example, a computer program that learns to play checkers might improve its performance as measured by its ability to win at the class of tasks involving playing checkers games, through experience obtained by playing games against itself.

Well defined learning problems

In general, to have a well-defined learning problem, we must identity these three features: the class of **tasks**, the **measure of performance** to be improved, and the source of **experience**.

A checkers learning problem:

Task T: playing checkers

Performance measure P: percent of games won against opponents

Training experience E: playing practice games against itself

Handwriting recognition learning problem

Task T: Recognising and classifying handwritten words within images

Performance P: Percent of words correctly classified

Training experience E: A dataset of handwritten words with given classifications

A robot driving learning problem

Task T: Driving on highways using vision sensors

Performance measure P: Average distance traveled before an error

Training experience E: A sequence of images and steering commands recorded while observing a human driver

Email Spam Filtering

Task T: Classifying emails as spam or not

Performance Measure P: The fraction of emails accurately classified as spam or not spam

Training experience E – Observing you label emails as spam or not spam

What is machine learning?

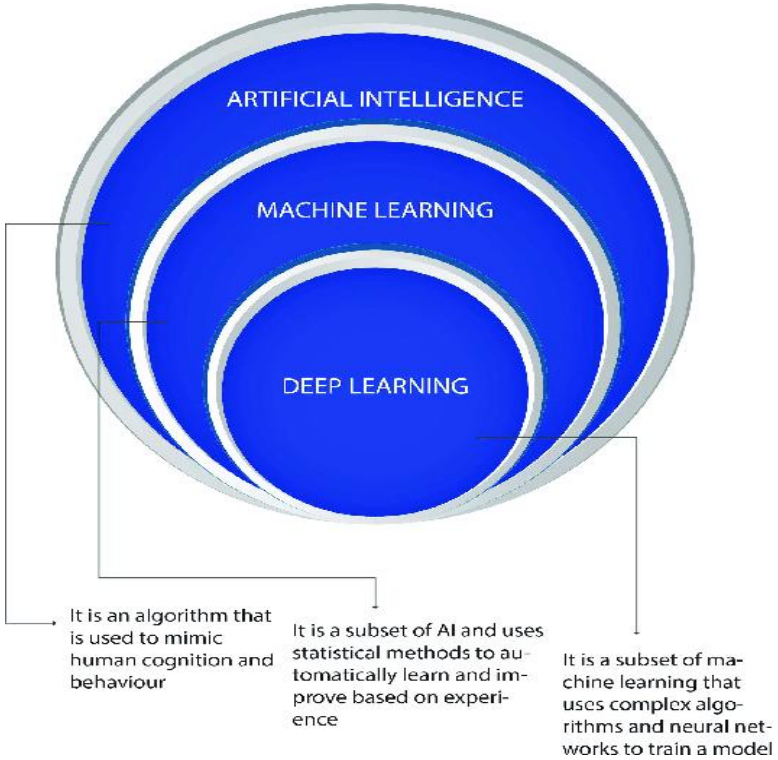
A computer program which **learns from experience** is called a machine learning program or simply a learning program. Such a program is sometimes also referred to as a learner.

Arthur Samuel, an early American leader in the field of computer gaming and artificial intelligence, coined the term "Machine Learning" in 1959 while at IBM.

He defined machine learning as "*the field of study that gives computers the ability to learn* ***without being explicitly programmed.***"

Machine Learning is defined as a discipline of Artificial Intelligence that provides machines the ability to automatically ***learn*** from data and past experiences to ***identify patterns*** and ***make predictions*** with ***minimal human intervention.***

Artificial Intelligence vs Machine Learning vs Deep Learning

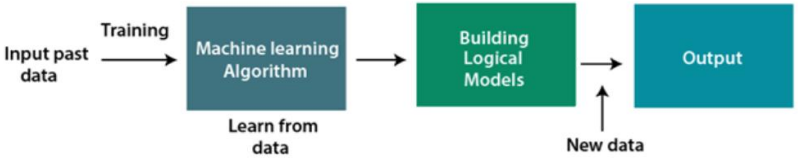


Artificial Intelligence vs Machine Learning vs Deep Learning

- **Artificial Intelligence** in general refers to the simulation of human intelligence processes by machines
- AI enables machine to think, without human intervention machine will be able to take own decision.
- **Machine learning** focuses on developing models and algorithms that let computers learn from **data** and improve from previous **experience without being explicitly programmed** for every task.
- **Deep learning** is a subset of machine learning in which multi-layered neural networks learn from vast amount of data.
- The **main difference** between ML and DL technologies is of presentation of data. ML uses structured and unstructured data for learning, while deep learning uses neural networks for learning models.

Designing a Learning System

- When we fed the Training Data to Machine Learning Algorithm, this algorithm will produce a **mathematical model/logical model.**
- With the help of the mathematical model, the machine will make a prediction and take a decision without being explicitly programmed.
- Also, during training data, the more machine will work with it the more it will get experience and the more efficient result is produced.



Example : In Driverless Car, the training data is fed to Algorithm like how to Drive Car in Highway, Busy and Narrow Street with factors like speed limit, parking, stop at signal etc. After that, a Logical and Mathematical model is created on the basis of that and after that, the car will work according to the logical model. Also, the more data the data is fed the more efficient output is produced.

Steps for Designing a learning system

- Choosing the Training Experience
- Choosing the Target Function
- Choosing a Representation for the Target Function
- Choosing a Function Approximation Algorithm
- The Final Design

1. Choosing the Training Experience

- The first design choice is to choose the **type of training experience** from which our system will learn.
- The type of training experience available can have a significant impact on success or failure of the learner.

<u>There are three key attributes which impact on success or failure of the learner</u>

1. Whether the training experience provides **direct** or **indirect** feedback regarding the choices made by the performance system.
2. The degree to which the learner controls the sequence of training examples
3. How well it represents the distribution of examples over which the final system performance P must be measured

1. Whether the training experience provides direct or indirect feedback regarding the choices made by the performance system.

For example, in checkers game:

- In learning to play checkers, the system might learn from **direct training examples** consisting of individual *checkers board states and the correct move for each*.

- **Indirect training examples** consisting of the *move sequences and final outcomes of various games played*.

    - The information about the correctness of specific moves early in the game must be inferred indirectly from the fact that the game was eventually won or lost.

    - Here the learner faces an additional problem of ***credit assignment***, or determining the degree to which each move in the sequence deserves credit or blame for the final outcome.

    - Credit assignment can be a particularly difficult problem because the game can be lost even when early moves are optimal, if these are followed later by poor moves.

    - Hence, learning from direct training feedback is typically easier than learning from indirect feedback.

2. Degree to which the learner controls the sequence of training examples

For example, in checkers game:

- The learner might depends on the trainer to select informative board states and to provide the correct move for each.
- Alternatively, the learner might itself propose board states that it finds particularly confusing and ask the teacher for the correct move.
- The learner may have complete control over both the board states and (indirect) training classifications, as it does when it learns by playing against itself with no teacher present.

Notice in this last case the learner may choose between experimenting with novel board states that it has not yet considered, or honing (sharpen) its skill by playing minor variations of lines of play it currently finds most promising.

3. How well training experience represents *the distribution of examples* over which the final system performance P must be measured.

Generally, Learning is most reliable when the training examples follow a distribution similar to that of future test examples.

For example, in checkers game:

- In checkers learning scenario, the performance metric P is the percent of games the system wins in the world tournament.

- If its training experience E consists only of games played against itself, there is an danger that this training experience might not be fully representative of the distribution of situations over which it will later be tested. For example, the learner might never encounter certain crucial board states that are very likely to be played by the human checkers champion.

- It is necessary to learn from a distribution of examples that is somewhat different from those on which the final system will be evaluated. Such situations are problematic because <u>mastery of one distribution of examples will not necessary lead to strong performance over some other distribution</u>.

2. Choosing the Target Function

The next design choice is to determine exactly what **type of knowledge will be learned** and how this will be used by the performance program.

For example, in checkers game:

- Lets begin with a checkers-playing program that can generate the legal moves from any board state.
- The program needs only to **learn how to choose the best move** from among these legal moves.
- Given this setting where we must learn to choose among the legal moves, the most obvious choice for the type of information to be learned is a **program, or function**, that chooses the best move for any given board state.

1. Let ***ChooseMove*** be the target function and the notation is $ChooseMove : B \rightarrow M$

   where **B** = set of all legal board states, **M** = set of legal moves. ChooseMove function accepts as input any board from the set of legal board states B and produces as output some move from the set of legal moves M.

   ChooseMove function will turn out to be very *difficult* to learn given the kind of *indirect training experience* available to our system

2. In alternative target function is an ***evaluation function*** that assigns a ***numerical score*** to any given board state

Let the target function V and the notation $V : B \rightarrow R$ which denote that V maps any legal board state from the set B to some real value

We intend for this **target function V** to assign **higher scores to better board states**. If the system can successfully learn such a target function V, then it can easily use it to select the best move from any current board position.

Let us define the target value V(b) for an arbitrary board state b in B, as follows:

1. if b is a final board state that is won, then V(b) = 100
2. if b is a final board state that is lost, then V(b) = -100
3. if b is a final board state that is drawn, then V(b) = 0
4. if b is a not a final state in the game, then V(b) = V(b'), where b' is the best final board state that can be achieved starting from b and playing optimally until the end of the game

To know V(b) in case 4, we'd have to search ***all the way to the end of the game*** *(optimal play).That's too slow and not realistic.*

So, we say this definition is **non-operational** (good in theory, but can't be used directly).

The program tries to find an **operational approximation** of V (a version that works fast enough to be used while playing).

3. Choosing a Representation for the Target Function

Now that we have specified the ideal target function V, we must choose a representation that the learning program will use to describe the function c that it will learn.

Different Ways to Represent V

We could represent it in many ways, for example:

1. **Big Table** → Store the score of *every possible board*. (Not practical → too many boards!)

2. **Rules** → Write "if-then" rules, like "if more kings, then higher score."

3. **Polynomial Function** → Use a math formula based on board features.

4. **Neural Network** → A flexible model that can learn complex patterns (but needs a lot of training data).

⚖️ **Trade-off:**

- A **complex representation** (like neural networks) can approximate V very well but needs a lot of data.

- A **simple representation** is easier to train but may not capture all details.

Lets choose simple representation, we'll use a **linear function of features** of the board.

We define some important features:

- x1: Number of black pieces
- x2: Number of red pieces
- x3: Number of black kings
- x4: Number of red kings
- x5: Number of black pieces in danger
- x6: Number of red pieces in danger

Then, the program estimates board value as:

$$\hat{V}(b) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 + w_6x_6$$

Here:

- **w0 ... w6** = weights (numbers learned by the program).
- These weights tell how important each feature is.
  - Example: If w3 (black kings) is very positive, it means "having black kings is very good."

Partial design of a checkers learning program:

- Task T: playing checkers
- Performance measure P: percent of games won in the world tournament
- Training experience E: games played against itself
- *Target function:* $V: Board \rightarrow \Re$
- *Target function representation*

$$\hat{V}(b) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_5 + w_6 x_6$$

4. Choosing a Function Approximation Algorithm

- In order to learn the target function V we require a set of training examples, each describing a specific board state b and training value $V_{train}(b)$ for b.
- Each training example is an ordered pair of the form $(b, V_{train}(b))$. B is the board state and $V_{train}(b)$ is the score of that board

Example:

If the board shows that **black has already won**, then

- $x_2=0$ (red has no pieces left)
- So $V_{train}(b)=+100V$

Training Example $\langle\langle x_1 = 3, x_2 = 0, x_3 = 1, x_4 = 0, x_5 = 0, x_6 = 0\rangle, +100\rangle$

So, Function Approximation Procedure will be described as—

1. Deriving training examples from indirect training experience available to the learner.
2. Adjust the weight w_i to best fit these training examples

4.1. Estimating training values

A simple approach for estimating training values for intermediate board states is to assign the training value of $V_{train}(b)$ for any intermediate board state b to be $\hat{V}$(Successor(b))

Where ,$\hat{V}$ is the learner's current approximation to V
Successor(b) denotes the next board state following b for which it is again the program's turn to move

Rule for estimating training values

$$V_{train}(b) \leftarrow \hat{V}(\text{Successor}(b))$$

This way, the program uses its **current guess** of the next state's value to estimate the current board's value.

Over time, as the program keeps refining, this "bootstrapping" process improves and can actually converge to good estimates.

4.2. Adjusting the weights

Specify the learning algorithm for choosing the weights w_i to best fit the set of training examples $\{(b, V_{train}(b))\}$

A first step is to define what we mean by the bestfit to the training data.

- One common approach is to define the best hypothesis, or set of weights, as that which minimizes the squared error E between the training values and the values predicted by the hypothesis.

$$E \equiv \sum_{\langle b, V_{train}(b)\rangle \in \text{ training examples}} (V_{train}(b) - \hat{V}(b))^2$$

- Several algorithms are known for finding weights of a linear function that minimize E.

This is the simple update rule to adjust weights gradually:

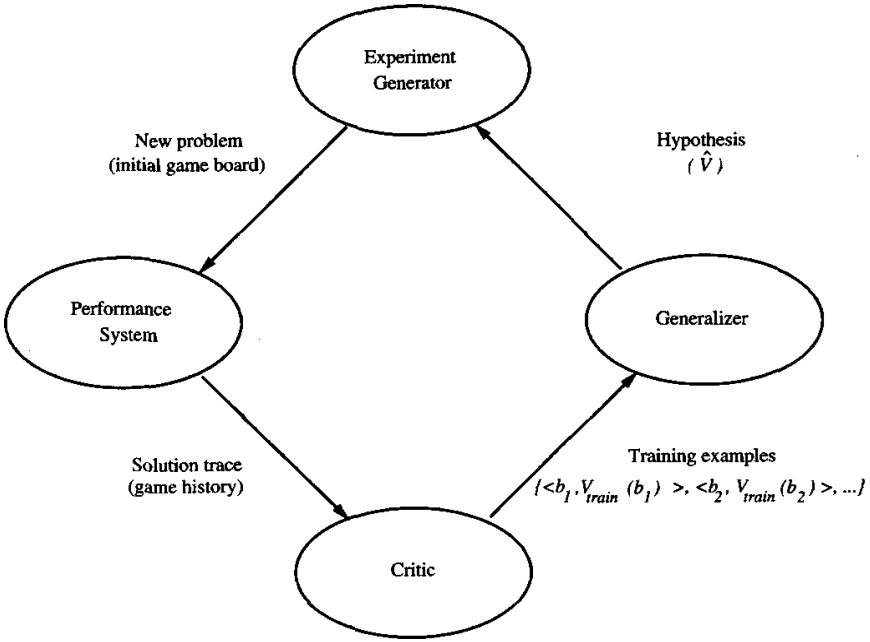
For each training example:

1. Calculate the predicted value $\hat{V}(b)$
2. Update each weight w_i as: $w_i \leftarrow w_i + \eta\,(V_{train}(b) - \hat{V}(b)) \cdot x_i$

- η = small constant (learning rate, e.g., 0.1).
- $(V_{\text{train}}(b) - f(b))$ = error.
- x_i = the feature value (like number of red pieces).

5. The Final Design

The final design of checkers learning system can be described by four distinct program modules that represent the central components in many learning systems



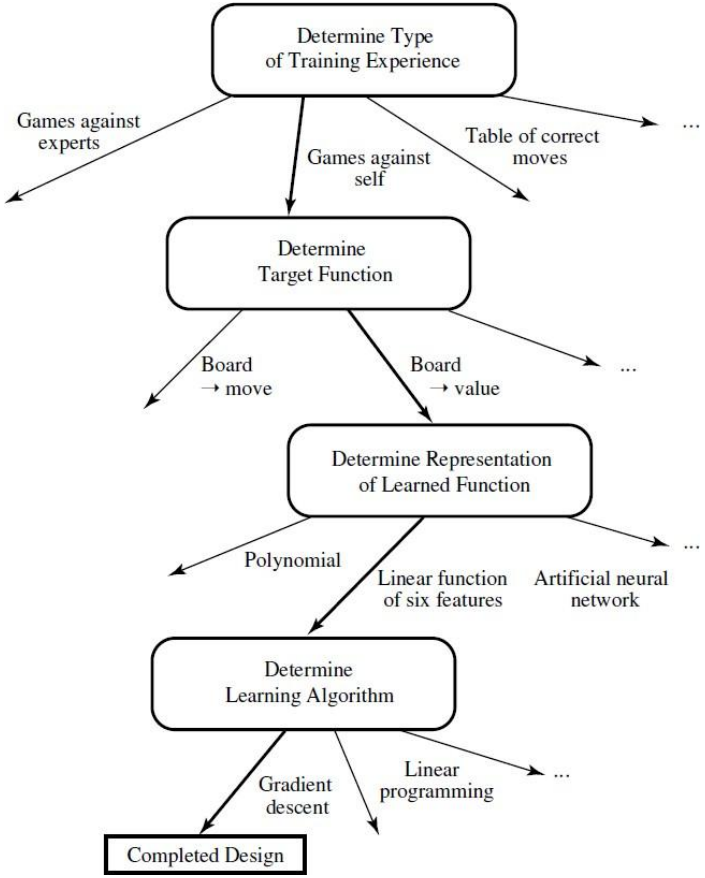
1. ***The Performance System*** is the module that must solve the given performance task by using the learned target function(s).
It takes an instance of a new problem (new game) as input and produces a trace of its solution (game history) as output.
In checkers game, the strategy used by the Performance System to select its next move at each step is determined by the learned $\hat{V}$ evaluation function. Therefore, we expect its performance to improve as this evaluation function becomes increasingly accurate.

2. ***The Critic*** takes as input the history or trace of the game and produces as output a set of training examples of the target function. As shown in the diagram, each training example in this case corresponds to some game state in the trace, along with an estimate V_{train} of the target function value for this example.

3. ***The Generalizer*** takes as input the training examples and produces an output hypothesis that is its estimate of the target function.
It generalizes from the specific training examples, hypothesizing a general function that covers these examples and other cases beyond the training examples.
In our example, the Generalizer corresponds to the LMS algorithm, and the output hypothesis is the function $\hat{V}$ described by the learned weights $w_0, \ldots, W_6$.

4. ***The Experiment Generator*** takes as input the current hypothesis and outputs a new problem (i.e., initial board state) for the Performance System to explore. Its role is to pick new practice problems that will maximize the learning rate of the overall system.
In our example, the Experiment Generator always proposes the same initial game board to begin a new game.

The sequence of design choices made for the checkers program is summarized in below figure



History of Machine Learning

- **Year 1950** : Alan Turing developed the Turing Test during this year.
- **Year 1957 : Perceptron -** The first ever Neural Network
- **Year 1960** : MIT developed a Natural Language Processing program to act as a therapist. The program was called ELIZA.
- **Year 1967** : The advent of Nearest Neighbor algorithm, very prominently used in Search and Approximation
- **Year 1970** : Backpropagation takes shape. Backpropagation is a set of algorithms used extensively in Deep Learning.
- **Year 1980** : Kunihiko Fukushima successfully built a multilayered Neural Network called ANN.
- **Year 1981** : Explanation Based Learning
- **Year 1989** : Reinforcement Learning is finally realized. Q-Learning algorithm.
- **Year 2009** : ImageNet
- **Year 2010** : Google Brain and Facebook's DeepFace
- Year 2022 : ChatGPT **Chat Generative Pre-trained Transformer**

https://www.zeolearn.com/magazine/what-is-machine-learning

Through the decades after the 1940s, the evolution of machine learning includes some of the more notable developments:

- Pioneers named Turing, Samuel, McCarthy, Minsky, Edmonds and Newell dotted the machine learning landscape in the 1950s, when the **Turing Test**, the first **artificial neural network**, and the terms *artificial intelligence* and *machine learning* were conceived.
- **Eliza**, the first chatbot; Shakey, the first mobile intelligent robot; Stanford Cart, a video-controlled remote vehicle; and the foundations of deep learning were developed in the 1960s.
- Programs that recognize patterns and handwritten characters, solve problems based on natural selection, seek appropriate actions, create rules to discard unimportant information, and learn like a baby learns to pronounce words highlighted the 1970s and 1980s.

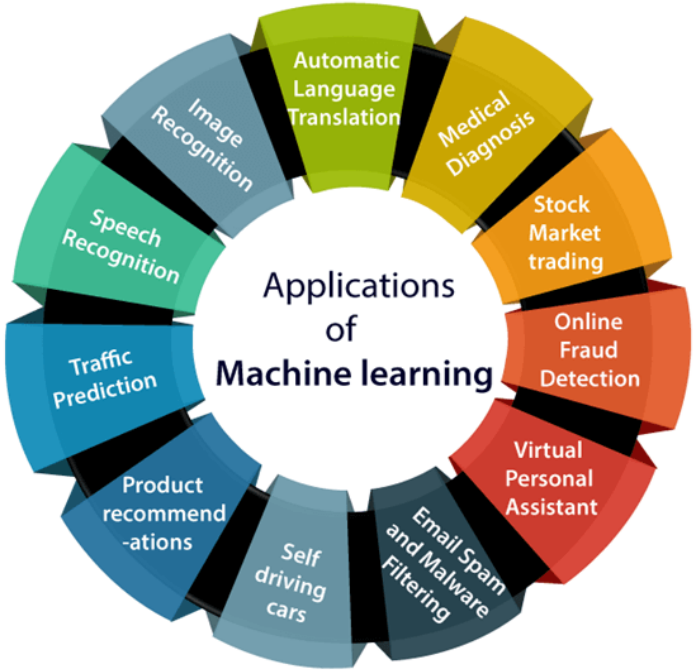
- Programs capable of playing backgammon and chess threatened the domains of top-tier backgammon players and the reigning world chess champion in the 1990s.
- In 2011, IBM's Watson won Jeopardy, a quiz show where it had to solve complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.
- Personal assistants, generative adversarial networks, facial recognition, deepfakes, motion sensing, autonomous vehicles, multimodal chatbot interfaces to LLMs, and content and image creation through the democratization of AI tools have emerged so far in the 2000s.
- In Year 2016 DeepMind's AlphaGo secured victory over the esteemed Go player Lee Sedol in Seoul, South Korea.
- In Year 2022 OpenAI launched ChatGPT, offering a chat-oriented interface to its GPT-3.5 LLM.

Importance of Machine Learning

- ***Machine learning is important because it gives enterprises a view of trends in customer behavior and operational business patterns, as well as supports the development of new products.***

- Many of today's leading companies, such as **Facebook, Google,** and **Uber**, make machine learning a central part of their operations. Machine learning has become a significant competitive differentiator for many companies.

Applications of Machine Learning



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1.Image Recognition:

- Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc.
- The popular use case of image recognition and face detection is, **Automatic friend tagging suggestion**.
- Facebook provides us a feature of auto friend tagging suggestion. Whenever we upload a photo with our Facebook friends, then we automatically get a tagging suggestion with name, and the technology behind this is machine learning's **face detection** and **recognition algorithm**.

2. Speech Recognition

- While using Google, we get an option of "Search by voice," it comes under speech recognition, and it's a popular application of machine learning.
- Speech recognition is a process of converting voice instructions into text, and it is also known as "Speech to text", or "Computer speech recognition." At present, machine learning algorithms are widely used by various applications of speech recognition. Google assistant, Siri, Cortana, and Alexa are using speech recognition technology to follow the voice instructions.

3. Traffic prediction:

- If we want to visit a new place, we take help of Google Maps, which shows us the correct path with the shortest route and predicts the traffic conditions.
- It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways

Real Time location of the vehicle form Google Map app and sensors

Average time has taken on past days at the same time.

- Everyone who is using Google Map is helping this app to make it better. It takes information from the user and sends back to its database to improve the performance.

4. Product recommendations:

- Machine learning is widely used by various e-commerce and entertainment companies such as Amazon, Netflix, etc., for product recommendation to the user. Whenever we search for some product on Amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.
- Google understands the user interest using various machine learning algorithms and suggests the product as per customer interest.
- As similar, when we use Netflix, we find some recommendations for entertainment series, movies, etc., and this is also done with the help of machine learning.

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5. Self-driving cars:

- One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars.
- Tesla, the most popular car manufacturing company is working on self-driving car. It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering:

- Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning.
- Some machine learning algorithms such as Multi-Layer Perceptron, Decision tree, and Naïve Bayes classifier are used for email spam filtering and malware detection.

7.Online Fraud Detection:

- Machine learning is making our online transaction safe and secure by detecting fraud transaction. Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place such as fake accounts, fake ids, and steal money in the middle of a transaction. So to detect this, Feed Forward Neural network helps us by checking whether it is a genuine transaction or a fraud transaction.
- For each genuine transaction, the output is converted into some hash values, and these values become the input for the next round. For each genuine transaction, there is a specific pattern which gets change for the fraud transaction hence, it detects it and makes our online transactions more secure.

8.Stock Market trading:

- Machine learning is widely used in stock market trading. In the stock market, there is always a risk of up and downs in shares, so for this machine learning's long short term memory neural network is used for the prediction of stock market trends.

9.Medical Diagnosis:

- In medical science, machine learning is used for **diseases diagnoses**. With this, medical technology is growing very fast and able to build **3D models that can predict the exact position of lesions in the brain**.
- It helps in finding brain tumours and other brain-related diseases easily.

10. Automatic Language Translation:

- Nowadays, if we visit a new place and we are not aware of the language then it is not a problem at all, as for this also machine learning helps us by converting the text into our known languages. Google's GNMT (Google Neural Machine Translation) provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.
- The technology behind the automatic translation is a sequence to sequence learning algorithm, which is used with image recognition and translates the text from one language to another language.

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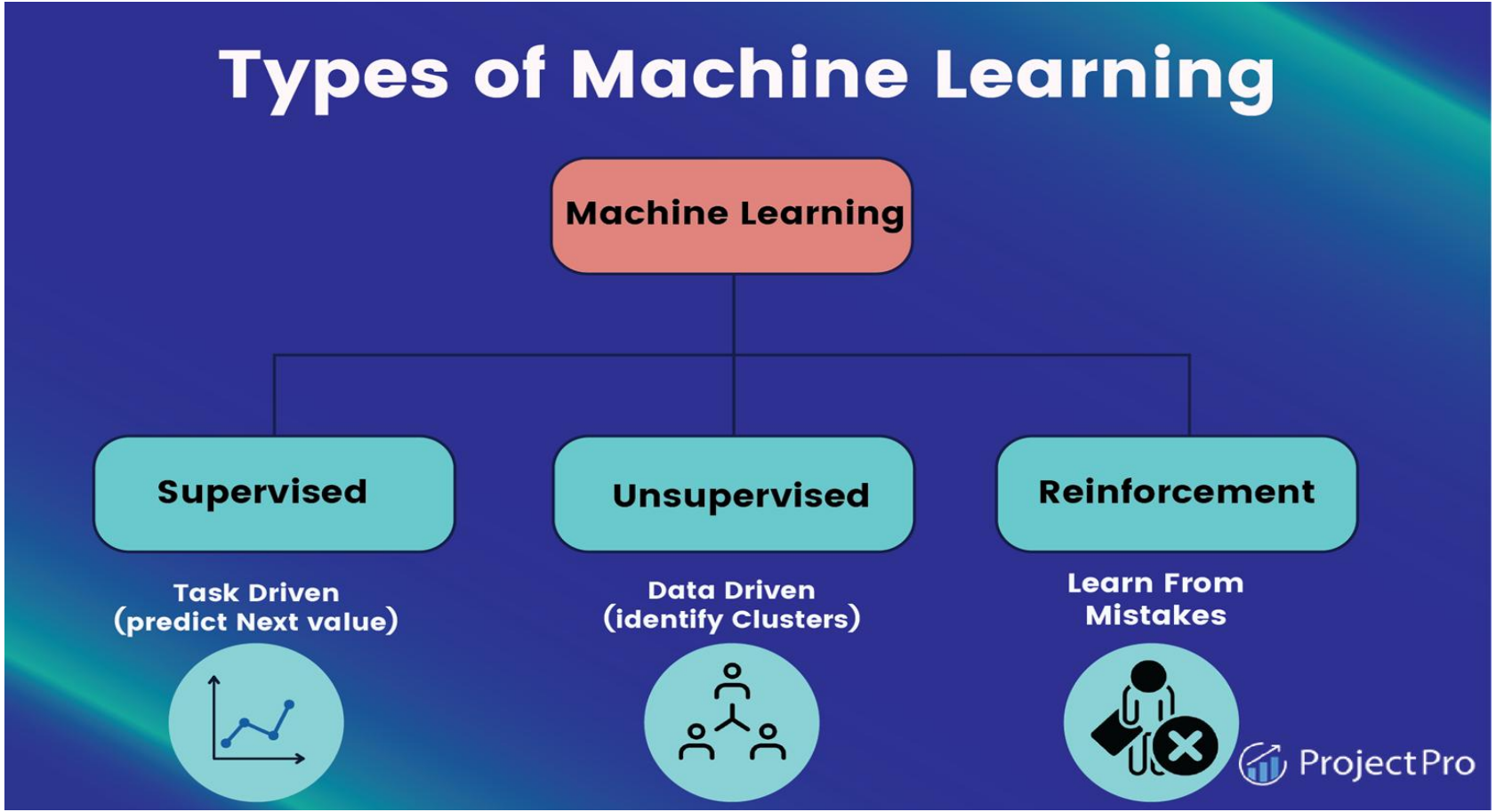
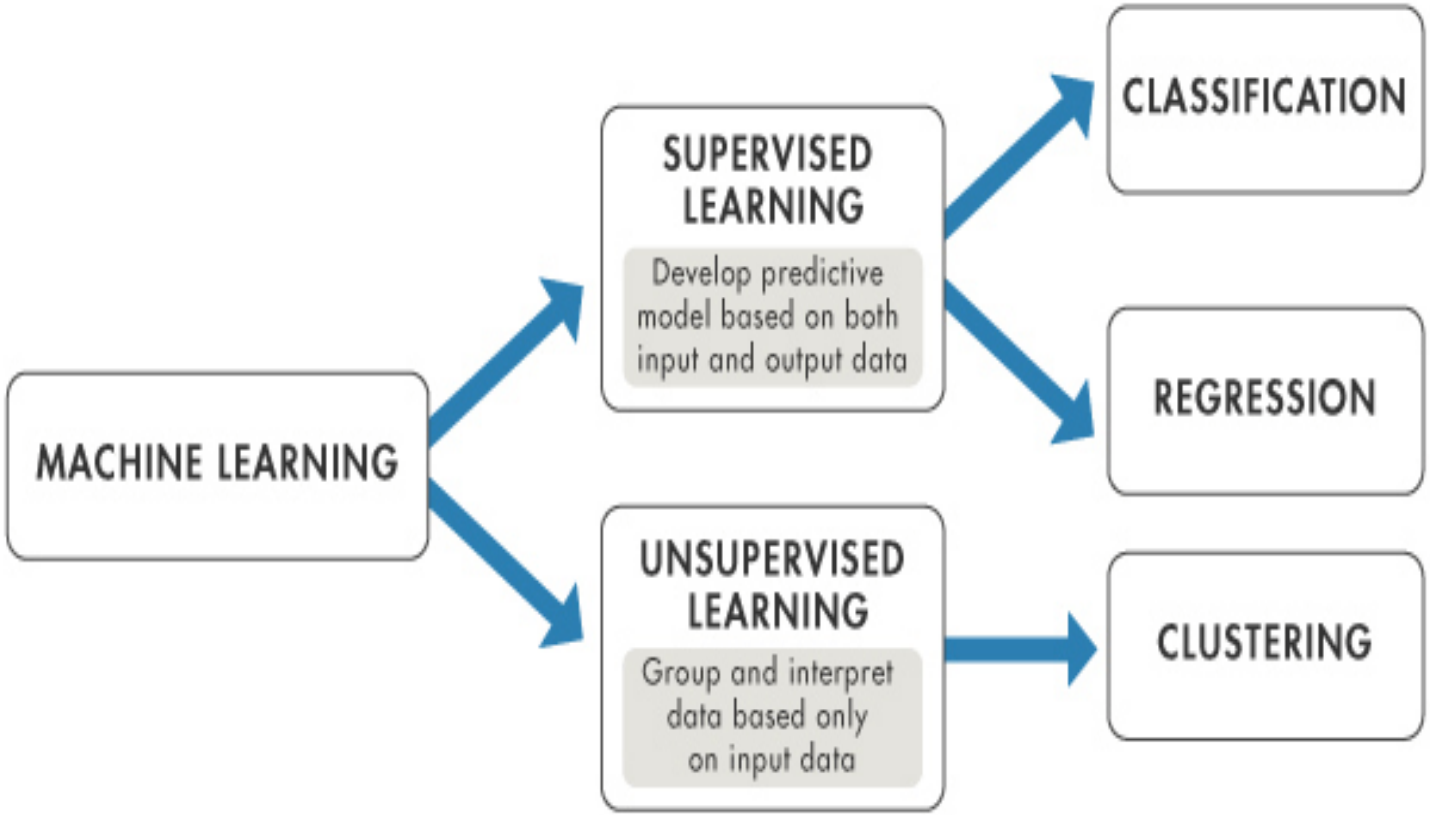
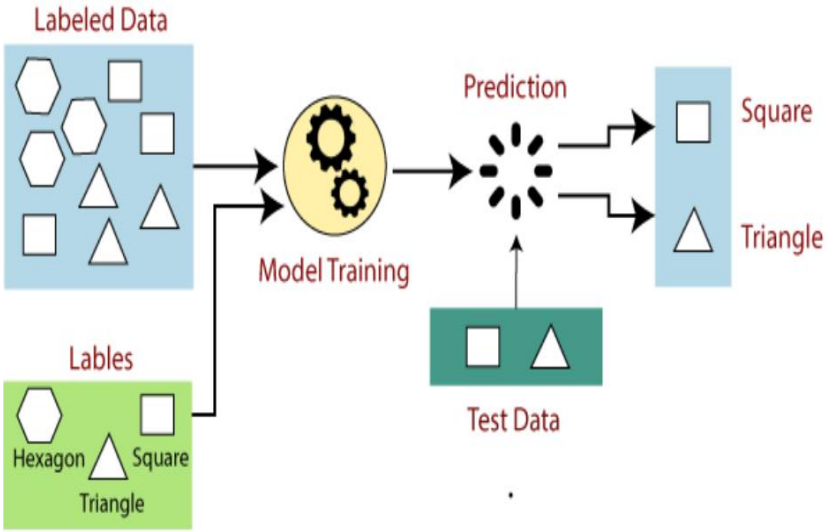

Types of Machine Learning
Machine Learning
Supervised
Unsupervised
Reinforcement
Task Driven
(predict Next value)
Data Driven
(identify Clusters)
Learn From
Mistakes
ProjectPro


MACHINE LEARNING
SUPERVISED LEARNING
Develop predictive model based on both input and output data
UNSUPERVISED LEARNING
Group and interpret data based only on input data
CLASSIFICATION
REGRESSION
CLUSTERING

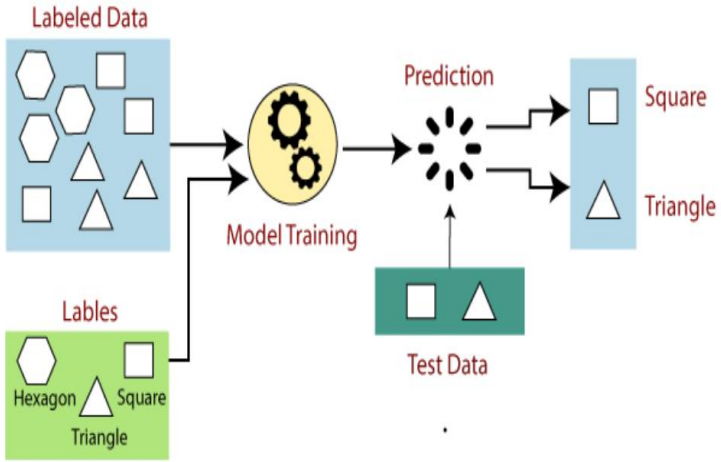
Supervised learning:

Supervised learning is the types of machine learning in **which machines are trained using well "labelled" training data,** and on basis of that data, machines predict the output. The labelled data means some input data is already tagged with the correct output.

ex-Image classification, fraud detection, filtering.



Supervised learning:



- Suppose we have a dataset of different types of shapes which includes square, rectangle, triangle, and Polygon. Now the first step is that we need to train the model for each shape.
- If the given shape has four sides, and all the sides are equal, then it will be labelled as a Square.
- If the given shape has three sides, then it will be labelled as a triangle.
- If the given shape has six equal sides then it will be labelled as hexagon.
- Now, after training, we test our model using the test set, and the task of the model is to identify the shape.
- The machine is already trained on all types of shapes, and when it finds a new shape, it classifies the shape on the bases of a number of sides, and predicts the output.

Types of supervised learning

1.**Regression**-

- Regression algorithms are used to solve regression problems in which there is a **linear relationship** between input and output variables.
- These are used to predict continuous output variables, such as market trends, weather prediction, etc.
- Some popular Regression algorithms are Simple Linear Regression Algorithm, Multivariate Regression Algorithm, Decision Tree Algorithm , Lasso Regression

2.**Classification-**

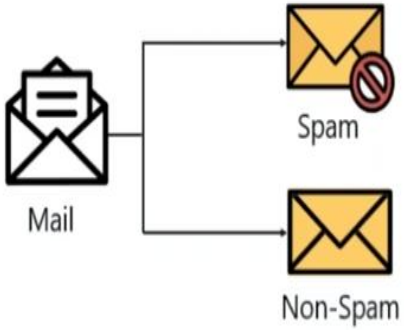
- A Classification problem is when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, red or blue, disease or no diseases etc.
- Some real-world examples of classification algorithms are Spam Detection, Email filtering, etc.
- Some popular classification algorithms are Random Forest Algorithm, Decision Tree Algorithm , Logistic Regression Algorithm, Support Vector Machine Algorithm

Regression Vs Classification

Classification

Classification is the task of predicting a discrete class label

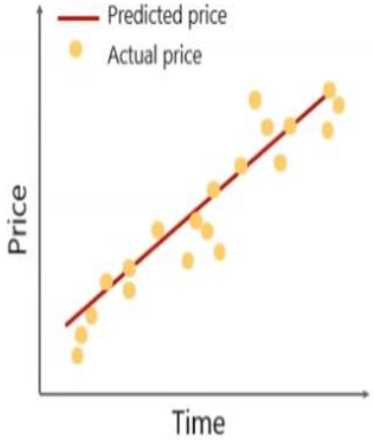
- In a classification problem data is classified into one of two or more classes
- A classification problem with two classes is called binary, more than two classes is called a multi-class classification



Regression

Regression is the task of predicting a continuous quantity

- A regression problem requires the prediction of a quantity
- A regression problem with multiple input variables is called a multivariate regression problem



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Advantages of Supervised learning:

- With the help of supervised learning, the model can predict the output on the basis of prior experiences.
- In supervised learning, we can have an exact idea about the classes of objects.
- Supervised learning model helps us to solve various real-world problems such as fraud detection, spam filtering, etc.

Disadvantages of supervised learning:

- Supervised learning models are not suitable for handling the complex tasks.
- Supervised learning cannot predict the correct output if the test data is different from the training dataset.
- Training required lots of computation time.
- In supervised learning, we need enough knowledge about the classes of object.

Unsupervised Machine Learning

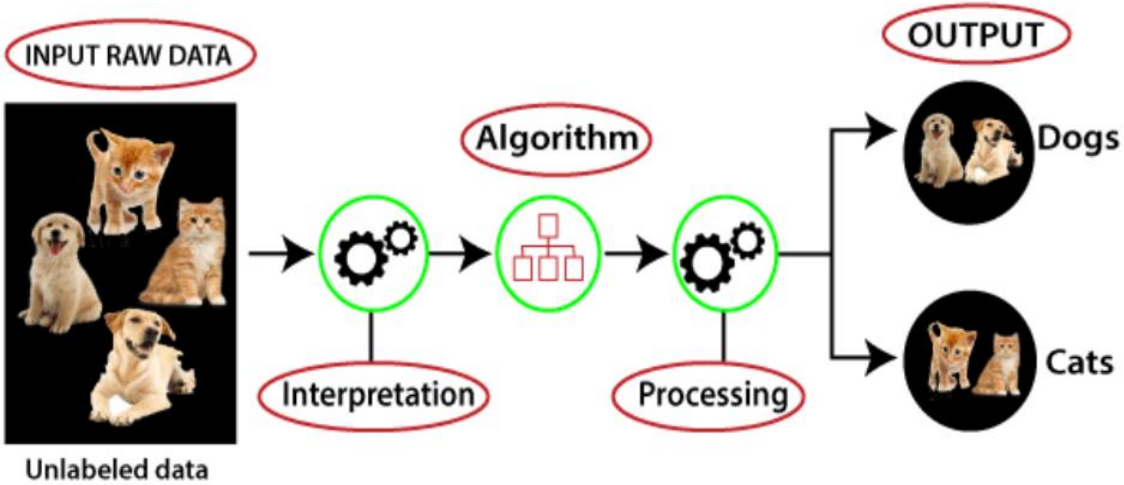
- As the name suggests, unsupervised learning is a machine learning technique in which models are not supervised using training dataset. Instead, models itself **find the hidden patterns and insights** from the given data.
- It can be defined as:

Unsupervised learning is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.

- The goal of unsupervised learning is to find the **underlying structure** of dataset, group that data according to similarities, and represent that dataset in a compressed format.

Example

- Suppose the unsupervised learning algorithm is given an input dataset containing images of different types of cats and dogs.
- The algorithm is never trained upon the given dataset, which means it does not have any idea about the features of the dataset.
- The task of the unsupervised learning algorithm is to identify the image features on their own.
- Unsupervised learning algorithm will perform this task by clustering the image dataset into the groups according to similarities between images.



Types of Unsupervised Learning

1. **Clustering**:
   - Clustering is a method of grouping the objects into clusters such that objects with **most similarities remains into a group** and has less or no similarities with the objects of another group.
   - Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.
   - An **example** of the clustering algorithm is grouping the customers by their purchasing behavior.
   - Some Popular Clustering algorithms are K-Means Clustering algorithm, Mean-shift algorithm, DBSCAN Algorithm, Principal Component Analysis, Independent Component Analysis

2. Association:

- An association rule is an unsupervised learning method which is used for **finding the relationships between variables** in the large dataset. It determines the set of items that occurs together in the dataset.
- Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.
- Some popular algorithms of Association rule learning are Apriori Algorithm, Eclat, FP-growth algorithm.

Advantages of Unsupervised Learning

- Unsupervised learning is used for more complex tasks as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.
- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.

Disadvantages of Unsupervised Learning

- Unsupervised learning is intrinsically more difficult than supervised learning as it does not have corresponding output.
- The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using **labeled data.**	Unsupervised learning algorithms are trained using **unlabeled data**.
The goal of supervised learning is to train the model so that it can predict the **output** when it is given new data.	The goal of unsupervised learning is to find the **hidden patterns** from the unknown dataset.
Supervised learning can be categorized in **Classification** and **Regression** problems.	Unsupervised Learning can be classified in **Clustering** and **Associations** problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an **accurate resul**t.	Unsupervised learning model may give **less accurate result** as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

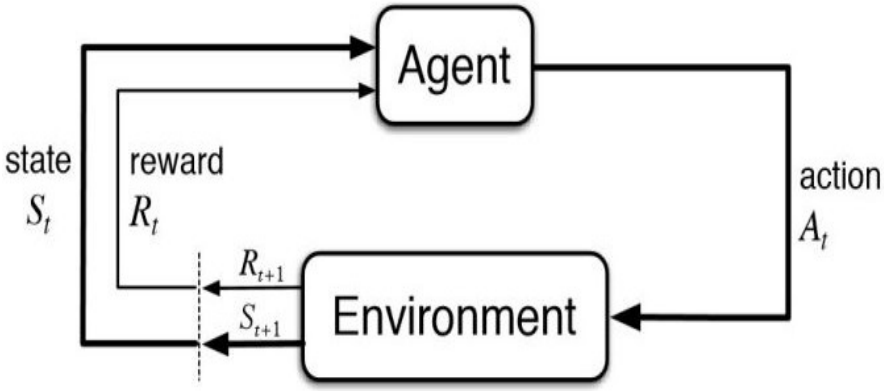
Reinforcement learning

- *Reinforcement learning is a feedback-based learning method, in which a learning agent gets a reward for each right action and gets a penalty for each wrong action.*
- The agent learns automatically with these feedbacks and improves its performance. In reinforcement learning, the agent interacts with the environment and explores it.
- The goal of an agent is to get the ***most* reward points**, and hence, it improves its performance.
- The robotic dog, which automatically learns the movement of his arms, is an example of Reinforcement learning.

- As compared to unsupervised learning, reinforcement learning is different in terms of goals. While the goal in **unsupervised learning** is to find similarities and differences between data points. RL is a technique that enables an agent to learn in an interactive **environment** by trial and error feedback from its own actions and experiences.
- Environment can be real world (robot) or simulated world (ex. Video game).
- In reinforcement learning the goal is to find a suitable action model that would **maximize the total cumulative reward** of the agent.
- The agent learns without intervention from a human by maximizing its reward and minimizing its penalty.
- RL can guide an agent on how to act in the real world (robot that vacuum your house, robot that learn to walk).
- Military is interested in this tech: RL agent can replace soldiers not just walk, but fight, diffuse bombs, make important decisions.

It is Inspired by human brain. We have eyes and nose as sensory organ but machine has sensors as sensory organ.

- **Agent-** an important component to process the information, thing that senses the environment.
- **Environment** – real world or simulated world that the agent lives in.
- **State**- different configuration of the environment that the agent can sense.
- **Actions**: are what an agent does in its environment. Notation (s,a,r).
- **Rewards**- positive or negative.



Types of Reinforcement learning

Reinforcement learning is categorized mainly into two types of methods/algorithms:

- **Positive Reinforcement Learning**: Positive reinforcement learning specifies increasing the tendency that the **required behavior would occur again by adding something.** It enhances the strength of the behavior of the agent and positively impacts it.

- **Negative Reinforcement Learning:** Negative reinforcement learning works exactly opposite to the positive RL. It increases the tendency that the specific behavior would occur again by **avoiding the negative condition.**

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Introduction of ML approach

a. ANN
b. Clustering
c. Reinforcement learning
d. Decision Tree learning
e. SVM
f. Genetic Approach
g. Bayesian Network

Issues in Machine Learning:

Machine learning, while powerful, comes with a variety of challenges and issues that can affect the development, deployment, and performance of models. Here are some of the key issues in machine learning:

1. Data Quality and Quantity

- **Insufficient Data:** Machine learning models *require large amounts of data to learn effectively. Insufficient data can lead to poor model performance.*
- **Data Imbalance:** In many real-world datasets, some classes are underrepresented, leading to *biased models that perform well on the majority class but poorly on the minority class.*
- **Noisy Data:** Data that *contains errors, outliers, or irrelevant features* can mislead the learning process, *resulting in inaccurate models.*
- **Data Preprocessing:** The process of cleaning, normalizing, and transforming data is crucial but can be complex and time-consuming.

2. Overfitting and Underfitting

- **Overfitting:** When a model learns the training data too well, including noise and outliers, it performs poorly on unseen data. Overfitting results in a model that is too complex for the problem.

- **Underfitting:** Occurs when a model is too simple to capture the underlying patterns in the data, leading to poor performance both on training and test data.

3. Model Interpretability

- **Black-Box Models:** Many advanced models, such as deep neural networks, are difficult to interpret. This lack of transparency can be problematic in applications where understanding the decision-making process is crucial, like healthcare or finance.

- **Explainability:** Providing explanations for predictions is increasingly important, especially in regulated industries. Balancing model accuracy with interpretability remains a significant challenge.

4. Computational Complexity

- **Training Time:** Some machine learning models, especially deep learning models, require significant computational resources and time to train, which can be a barrier for many practitioners.

- **Scalability:** As datasets grow in size, both in terms of samples and features, the computational requirements for training and deploying models increase dramatically.

- **Resource Constraints:** Deploying machine learning models in resource-limited environments, such as mobile devices or edge computing, requires careful optimization.

5. Generalization

- **Domain Adaptation:** Models trained on data from one domain may not generalize well to another domain, a problem known as domain shift. This issue is common when there are differences in the data distribution between the training and test datasets.

- **Transfer Learning:** While transfer learning aims to apply knowledge from one domain to another, it’s not always straightforward and can result in performance degradation if the domains are too different.

6. Ethical and Privacy Concerns

- **Data Privacy:** Machine learning models often require large amounts of personal data, raising concerns about data privacy and security. Ensuring compliance with regulations like GDPR is crucial.

- **Ethical Use:** The potential misuse of machine learning models, such as in surveillance or autonomous weapons, raises significant ethical questions.

DS VS ML

Data science and **machine learning** are closely connected but have different areas of focus. Here's a simplified comparison:

1. **What They Are:**
   - **Data Science:** Involves exploring, organizing, and analyzing data to discover useful information or patterns.
   - **Machine Learning:** A branch of data science that deals with creating systems that can learn from data and make decisions or predictions without being explicitly programmed.
2. **Tools Used:**
   - **Data Science:** Relies on tools like Excel, Python, R, SQL, and data visualization software to process and interpret data.
   - **Machine Learning:** Uses tools such as TensorFlow, Scikit-learn, and PyTorch to develop models that can learn and improve from data.

3. **Goals:**
   - **Data Science:** Aims to analyze and understand data to provide insights that guide better decisions.
   - **Machine Learning:** Focuses on building models that automatically predict outcomes or classify data based on patterns.
4. **Results:**
   - **Data Science:** Generates insights, reports, and visualizations that explain data trends and patterns.
   - **Machine Learning:** Produces models that can make automated predictions, such as recommending items or identifying images.
5. **Skills Required:**
   - **Data Science:** Involves skills in statistics, data manipulation, and visualization to interpret and explain data.
   - **Machine Learning:** Requires a strong understanding of algorithms, mathematics, and programming to create learning models.

🏦 **Example: Banking (Fraud Detection)**

- Data Science: Analyzes past transactions to find patterns of fraud (e.g., “Fraud cases often happen at midnight, from certain regions”).
- Machine Learning: Builds a fraud detection model that predicts in real-time whether a transaction is fraudulent.

🏥 **Example: Healthcare (Patient Records)**

- Data Science: Summarizes patient data (e.g., “Diabetes is most common in patients aged 40–60 in this dataset”).
- Machine Learning: Predicts whether a patient is at risk of developing diabetes based on their medical history and lifestyle data

Machine Learning	VS	Data Science
Machine Learning aims for automation and predictions		Data Science aims for extracting insights and supporting decision-making
Machine Learning focuses on creating predictive models		Data Science encompasses data analysis process
Required Understanding of Regression & Clustering		Works using Statistics and Data Visualization
Recommendation Systems, Fraud detection, Autonomous vehicles		Business intelligence and strategy, Market trend analysis, Customer segmentation